MAIS PAC

IPhO 02.13: Quantum Physics: Waves, Atoms and Spectra · Lecture + Problems

Quantum Physics: Waves, Atoms and Spectra

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. The failures of classical physics

By 1900 three experiments refused to fit classical physics:

- **Black-body radiation.** The spectrum of a hot object peaks at a wavelength that depends on temperature, and classical equipartition predicted infinite energy at short wavelengths — the "ultraviolet catastrophe".
- **The photoelectric effect.** Light above a threshold frequency ejects electrons from a metal instantly, however faint; below the threshold, no electrons at all, however bright. A wave theory predicts the opposite: weak light should take time, and bright light should always work.
- **Line spectra.** Atoms emit only certain sharp frequencies, not a continuous spectrum.

All three were resolved by one idea: energy is exchanged in discrete lumps.

2. Photons

Planck introduced quantization to fix the black-body spectrum; Einstein took it literally and proposed that light **is** composed of photons of energy

$$E = hf = (hc)/(\lambda),\ h = 6.63\times10^{(-34)}\ \{J,s\}. \tag{1}$$

It is often convenient to use hc = 1240 {eV,nm}, so that λ = 1240 {nm} corresponds to 1 {eV} — a mental shortcut worth having at your fingertips.

The photoelectric equation is then pure bookkeeping:

$$K_{(max)} = hf - \varphi, \tag{2}$$

where φ is the **work function** of the metal — the minimum energy needed to remove an electron. A photon either has enough energy or it does not: doubling the intensity doubles the **number** of photons but not the energy of each one.

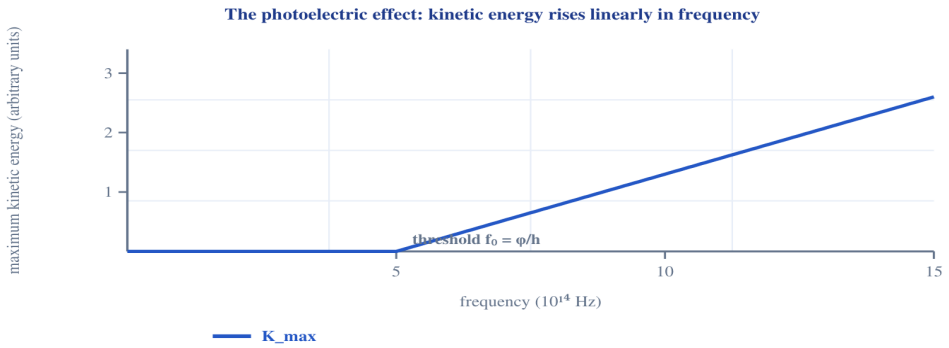


Figure. Below the threshold frequency nothing is emitted however bright the light; above it, only the frequency sets the energy.

Interactive experiment: The photoelectric effect — open the online handout to use it.

Consequences, all verified:

- A threshold frequency f_0 = φ/h, below which nothing happens.
- K_(max) rises linearly with f, with slope h — the standard way to measure Planck's constant.
- The photocurrent is proportional to intensity (above threshold), not the energy of each electron.
- Emission is essentially instantaneous.

The Compton effect confirmed the particle picture further: X-rays scattered from electrons change wavelength by

$$\Delta\lambda = (h)/(m_ec)(1-\cos\theta), \tag{3}$$

exactly as if a photon of momentum p = h/λ = E/c had bounced off a free electron like a billiard ball.

3. Matter waves

De Broglie proposed the reverse: if waves can be particles, particles can be waves, with

$$\lambda = (h)/(p) = (h)/(mv) \{or\} \lambda = (h)/(\sqrt{(2mK)}) \{ \text{ for kinetic energy } \} K. \quad (4)$$

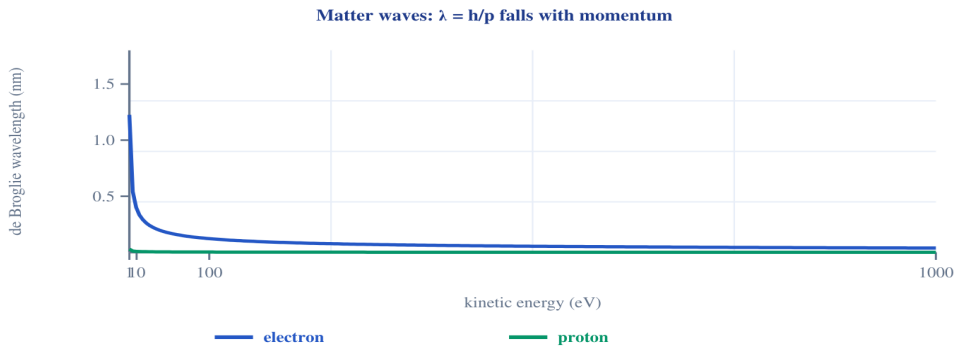


Figure. An electron at 100 eV has a wavelength of about 0.12 nm, comparable with atomic spacings.

These are not decorative: electron microscopes resolve atomic structure because 0.1 {keV} electrons have λ≈0.12 {nm}, comparable with the spacing of atoms, and electron diffraction from crystals shows the same patterns as X-rays.

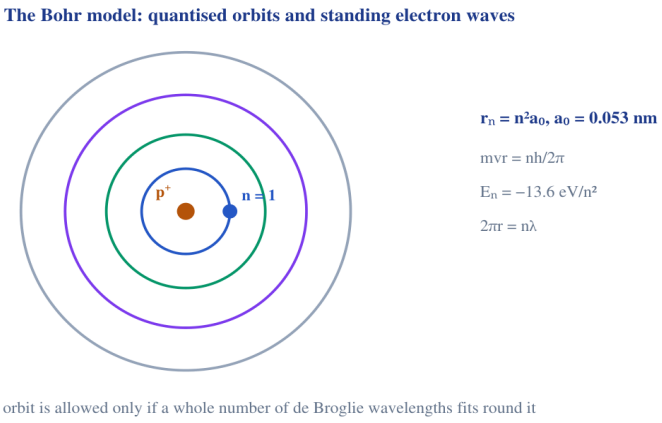


Figure 4. Bohr's assumption looked arbitrary in 1913, but it is exactly the condition that the electron forms a standing wave round the nucleus — the first hint that matter waves are real.

The Bohr model becomes natural in this language: an electron can orbit the nucleus only if a whole number of wavelengths fits round the orbit, 2π r = nλ. Combined with mvr = nh/(2π), this gives

$$r_n = n^2 a_0,\ E_n = -(13.6 \{eV\})/(n^2),\ a_0 = 0.053 \{nm\}. \quad (5)$$

4. Spectra

Downward transitions emit a photon of energy equal to the level difference,

$$hf = E_i - E_f,\ (1)/(\lambda) = R((1)/(n_f^2)-(1)/(n_i^2)),\ R = 1.097\times10^{7} \{m^{-1}\}. \quad (6)$$

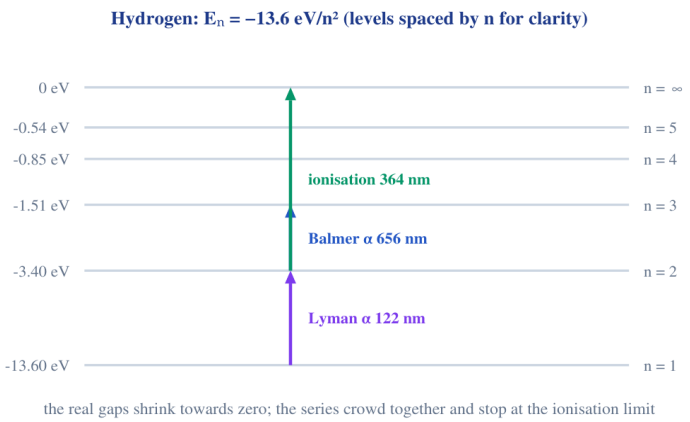


Figure 2. Every downward arrow is a photon of energy equal to the level difference, so hydrogen's spectrum is a set of sharp lines — a fingerprint of the atom.

The series have names: **Lyman** (n_f=1, ultraviolet), **Balmer** (n_f=2, visible — the familiar red 656 nm, blue-green 486 nm), **Paschen** (n_f=3, infrared). The same lines appear in emission (bright) and absorption (dark), and the pattern identifies the element.

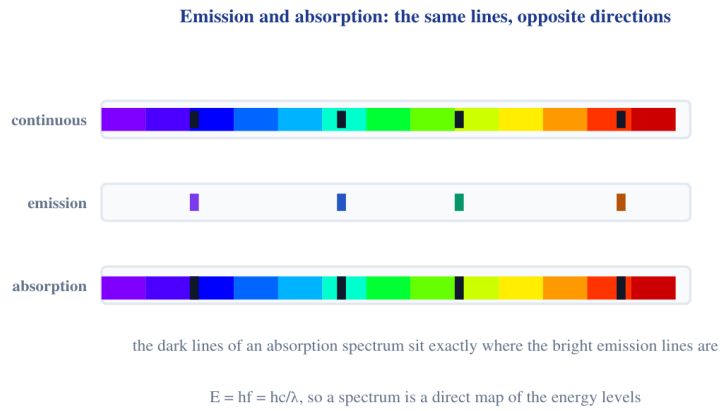


Figure 6. Astronomers read composition, temperature and motion from these lines: the pattern identifies the element, the brightness the abundance, and the Doppler shift the velocity.

5. The uncertainty principle and wave functions

If a particle is a wave, it cannot have both a sharp position and a sharp momentum, because a narrow pulse in space must be built from a wide spread of wavelengths:

$$\Delta x,\Delta p\ ()/(2),\ \Delta E,\Delta t\ ()/(2),\ =\ (h)/(2\pi). \tag{7}$$

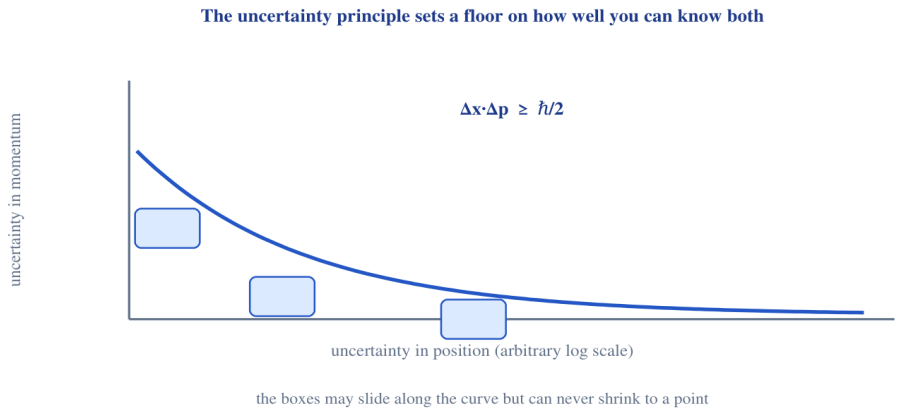


Figure 5. The relation is not about clumsy instruments: it is a property of waves. A wave packet that is narrow in space must contain a wide range of wavelengths, and hence a wide range of momenta.

This is not about imperfect instruments, and it is not about "disturbing" the particle. It is a statement about what a wave **is**. Its consequences are concrete:

- **Zero-point energy:** a particle in a box can never be at rest; its lowest energy is finite.
- **Confinement:** an electron held within a region of size a has a minimum momentum about /a, hence a minimum kinetic energy that grows as a shrinks — the reason atoms have a definite size and white dwarfs do not collapse at zero pressure.
- **Virtual particles and line widths:** the energy–time relation explains why short-lived states have broad spectral lines.

The full theory replaces orbits with a **wave function** whose squared modulus gives the probability of finding the particle. For the hydrogen atom the allowed wave functions give the familiar orbitals, and for tunnelling through a barrier of height U the transmission probability depends exponentially on the barrier width:

$$T \approx e^{(-2\ d),} = (\sqrt{(2m(U-E))})/(). \tag{8}$$

The quantum checklist. Convert everything to electronvolts early: hc = 1240 {eV,nm} does most of the arithmetic → for photons, E = hc/λ and p = E/c → for photoelectric problems, check the threshold **before** computing anything → for de Broglie wavelengths, use p = h/λ and K = p²/2m (or the relativistic form for high energies) → for spectra, an energy-level diagram turns every problem into a subtraction → never use λ f = c for an electron: that is for light only.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — A photon's energy. Find the energy, in joules and in electronvolts, and the momentum of a photon of (a) green light, λ = 550 {nm}; (b) ultraviolet light, λ = 200 {nm}; (c) an X-ray of λ = 0.10 {nm}.

Worked solution →

[1] Problem 2 — The photoelectric effect. Light of wavelength 400 {nm} falls on a metal with work function 2.30 {eV}. (a) Find the photon energy. (b) Find the maximum kinetic energy of the emitted electrons and the stopping voltage. (c) Find the threshold wavelength.

Worked solution →

[2] Problem 3 — Measuring Planck's constant. For a certain metal, the stopping voltage is 0.50 {V} at λ = 400 {nm} and 1.20 {V} at λ = 300 {nm}. (a) Find h from these two points. (b) Find the work function. (c) Find the threshold wavelength, and explain why the slope of the graph does not depend on the metal.

Worked solution →

[2] Problem 4 — De Broglie wavelength.

Find (a) the de Broglie wavelength of an electron accelerated through 100 {V}. (b) Find it for an electron at 100 {keV}, using the relativistic momentum. (c) Find the speed of a 1.0 {g} bead with the same wavelength as the 100 {V} electron, and explain why we never notice such wavelengths.

Worked solution →

[2] Problem 5 — Hydrogen lines. Use the Rydberg formula with R = 1.097×10^7 {m^(-1)}. (a) Find the wavelength of the n=3→2 transition. (b) Find the shortest wavelength in the Balmer series. (c) Find the energy of the photon for the n=2→1 transition and the recoil speed of the hydrogen atom.

Worked solution →

[2] Problem 6 — The Bohr model. For the hydrogen atom: (a) show that the orbital radius is $r_n = n^2 a_0$ and evaluate a_0 from $m_e v r = nh/2\pi$. (b) Find the speed of the electron in the ground state as a fraction of c. (c) Find the energy of the n=2 level and check it against $-13.6/n^2$.

Worked solution →

[2] Problem 7 — Compton scattering. An X-ray photon of wavelength 0.0500 {nm} is scattered through 90° by a free electron. (a) Find the wavelength of the scattered photon. (b) Find the kinetic energy given to the electron. (c) Explain why the shift is unobservable for visible light.

Worked solution →

[3] Problem 8 — The uncertainty principle. An electron is confined to a region of width 0.10 {nm} (about the size of an atom). (a) Estimate the minimum uncertainty in its momentum. (b) Estimate its minimum kinetic energy and compare it with the 13.6 {eV} binding energy of hydrogen. (c) Repeat for a proton confined to a nucleus of width 1.0 {fm} and comment.

Worked solution →

[3] Problem 9 — A photon and an electron with the same energy. A photon and an electron each have an energy of 2.0 {keV}. (a) Find the photon's wavelength and momentum. (b) Find the electron's momentum and wavelength (non-relativistic is adequate here — check it). (c) Explain why the two wavelengths differ, and which particle is more useful for imaging atoms.

Worked solution →

[3] Problem 10 — An electron in a box. Model a molecule as a one-dimensional box of length L = 1.0 {nm} with infinitely high walls, so that the electron's de Broglie wave fits a whole number of half wavelengths. (a) Find the allowed wavelengths and show that $E_n = n^2h^2/(8mL^2)$. (b) Evaluate E_1 and E_2 in eV. (c) Find the wavelength of the photon emitted in the 2→1 transition and comment on where it lies in the spectrum.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

PHYSNET / modern physics overview

Short modules on quanta, spectra, relativity and nuclear physics written for students.